ROOT CAUSE ANALYSIS · INC-4821

Database pool misconfiguration caused checkout failures

Sev2 Final · version 1

1 Incident overview

Incident ID	INC-4821
Severity	Sev2
Incident date	3 Nov 2026
Incident commander	ravi.kumar
Affected services	payments-api, checkout-web, payments-db
Document status	Final
Version	1
Prepared by	Asha Rao
Date prepared	12 Nov 2026

2 Executive summary

A misconfigured database pool size in deploy 2291 caused checkout failures for approximately 4,200 orders, impacting 30% of transactions across the payments-api, checkout-web, and payments-db services. The root cause was a typo that set the database pool maximum to 5 instead of 50, a value that was not validated by the continuous integration pipeline. The incident was resolved within one hour and five minutes by rolling back deploy 2291. Follow-up actions include adding schema validation for pool settings in CI, alerting on database connection saturation above 80%, and automatically linking deploy events to the incident channel.

3 Impact

The incident resulted in a 30% checkout failure rate, affecting approximately 4,200 orders. This disruption impacted the payments-api, checkout-web, and payments-db services, causing checkout failures for many users. The event was classified as Sev2 and lasted from 14:05 to 15:10, with a total resolution time of 1 hour and 5 minutes.

Customer impact	checkout failures for many users
Scale of impact	30% (≈4,200 orders)
Business or SLA impact	no SLA breach; 37 support tickets
Data loss or exposure	none
Duration	1 hour 5 minutes

4 Timeline

Milestone	Time as recorded	Time (UTC)
Incident start	2026-11-03 14:05	2026-11-03 08:35 UTC
Detected	2026-11-03 14:06	2026-11-03 08:36 UTC
Mitigated	2026-11-03 14:40	2026-11-03 09:10 UTC
Resolved	2026-11-03 15:10	2026-11-03 09:40 UTC

Times were recorded in IST, taken as Asia/Kolkata.

Time to detect: **1 minute** Time to mitigate: **35 minutes** Time to resolve: **1 hour 5 minutes**

Detailed timeline

Time	UTC	Event	Source
2026-11-03 14:06	2026-11-03 08:36 UTC	Priya reports wall of 500s on payments-api causing checkout failures	priya.n
2026-11-03 14:07	2026-11-03 08:37 UTC	Priya notes that the issue started around 14:05	priya.n
2026-11-03 14:12	2026-11-03 08:42 UTC	PagerDuty alerts on payments-api 5xx rate >20%	pagerduty
2026-11-03 14:13	2026-11-03 08:43 UTC	Ravi takes incident control and calls it sev2	ravi.kumar
2026-11-03 14:15	2026-11-03 08:45 UTC	Ravi notes checkout-web also failing	ravi.kumar
2026-11-03 14:18	2026-11-03 08:48 UTC	Meera reports db connections maxed out, pool saturated	meera.i
2026-11-03 14:22	2026-11-03 08:52 UTC	Meera says deploy 2291 deployed at 14:03 with config change	meera.i
2026-11-03 14:25	2026-11-03 08:55 UTC	Ravi initiates rollback of deploy 2291	ravi.kumar
2026-11-03 14:40	2026-11-03 09:10 UTC	Ravi reports rollback completed, error rate dropping	ravi.kumar
2026-11-03 14:41	2026-11-03 09:11 UTC	Priya reports checkout success rate recovering	priya.n
2026-11-03 15:10	2026-11-03 09:40 UTC	Ravi reports all green for 30 min, closes incident, resolved at 15:10	ravi.kumar
2026-11-03 15:20	2026-11-03 09:50 UTC	Meera identifies root cause: config change set db pool max to 5 instead of 50	meera.i
2026-11-03 15:22	2026-11-03 09:52 UTC	Meera reports impact: 30% failures between 14:05-14:40, about 4,200 orders	meera.i
2026-11-03 15:25	2026-11-03 09:55 UTC	Ravi reports no SLA breach, inside error budget, 37 tickets	ravi.kumar
2026-11-03 15:30	2026-11-03 10:00 UTC	Ravi notes what went well: alert fast, rollback clean	ravi.kumar
2026-11-03 15:31	2026-11-03 10:01 UTC	Ravi notes what went badly: 10 min to connect deploy to errors, deploy channel noisy	ravi.kumar
2026-11-03 15:35	2026-11-03 10:05 UTC	Meera lists actions to prevent recurrence	meera.i

5 Detection

Detection of the incident occurred within one minute of the start, at 14:06, following user reports of 500 errors on the payments-api. A PagerDuty alert was generated during this period, although the high volume of noise in the deploy channel contributed to a delay in linking the deployment to the observed errors.

How it was detected	user reported 500 errors on payments-api
Monitoring gaps	deploy channel is too noisy, causing delay in linking deploy to errors

6 Response and resolution

The response was led by Ravi, who assumed incident command at 14:13 and classified the event as sev2. The team executed a rollback of deploy 2291, which was initiated at 14:25 and completed by 14:40, resulting in a drop in error rates and the recovery of checkout success. The incident was formally resolved at 15:10 after a period of stability, with a total time to mitigate of 35 minutes and a time to resolve of 1 hour 5 minutes.

Root cause identification followed the mitigation, with Meera determining at 15:20 that a configuration change in deploy 2291 had set the database pool maximum to 5 instead of 50. Although the deploy was linked to the errors, there was a 10-minute delay between the deployment at 14:03 and the identification of the deploy as the source of the issue at 14:22. The response was characterized by swift incident command and a clean rollback, with the team noting that the alerting was fast and the mitigation process was effective.

Mitigation steps	1. rolled back deploy 2291
Fix applied	rollback of deploy 2291
People and teams involved	• pri… • ravi.kumar • meera.i

7 Root cause analysis

The root cause of the incident was a typo in deploy 2291, which set the database pool maximum to 5 instead of the intended value of 50. This configuration error was compounded by the fact that the continuous integration process did not validate configuration values before deployment.

Trigger	config change in deploy 2291 that set db pool max to 5
Root cause	config change in deploy 2291 set db pool max to 5 instead of 50 (typo); CI doesn\'t validate config values
Contributing factors	• CI doesn\'t validate config values
5 Whys	*Not provided by the user (Asha Rao, 12 Nov 2026)*

8 Lessons learned

What went well

- alert fired fast and the rollback was clean

What went wrong

- took 10 min to connect the deploy to the errors due to noisy deploy channel

9 Corrective and preventive actions

#	Action	Type	Owner	Due date	Ticket
1	add schema validation for pool settings in config CI	—	meera	20 Nov	—
2	alert on db connection saturation > 80%	—	priya	27 Nov	—
3	link deploy events into the incident channel automatically	—	*Not provided by the user (Asha Rao, 12 Nov 2026)*	—	—

10 Related documents

No similar past incidents were found in this tool's records.

11 Information gaps

The details below were asked for while preparing this document and were not provided.

Detail	Section	Status	Asked of	Date
5 Whys	Root cause analysis	*Not provided by the user*	Asha Rao	12 Nov 2026
Owner of action: link deploy events into the incident channel automatically	Corrective and preventive actions	*Not provided by the user*	Asha Rao	12 Nov 2026

12 Appendix and revision history

Revision history

Version	Date	Prepared by	Status
1	12 Nov 2026	Asha Rao	Approved by Asha Rao

> How this document was produced: the RCA Assistant extracted the facts from notes supplied by Asha Rao, checked each one against a quote from those notes, asked for missing details, and drafted the narrative sections using openai/gpt-oss-120b, openai/gpt-oss-20b, qwen/qwen3.8-27b via api.groq.com. Layout, durations and the gaps table are generated by code. It was approved by Asha Rao. Generated 12 Nov 2026 08:42.

Source of each fact

Field	Value	Supporting quote	Source
Incident ID	INC-4821	“[14:13] ravi.kumar: I'll take IC. calling it sev2 for now. INC-4821”	Notes
Severity	Sev2	“[14:13] ravi.kumar: I'll take IC. calling it sev2 for now. INC-4821”	Notes

Field	Value	Supporting quote	Source
Incident date	2026-11-03	“#inc-4821-payments (channel export, 3 Nov 2026, all times IST)”	Notes
Incident commander	ravi.kumar	“[14:13] ravi.kumar: I'll take IC. calling it sev2 for now. INC-4821”	Notes
Customer impact	checkout failures for many users	“[14:06] priya.n: seeing a wall of 500s on payments-api, checkout is failing for a lot of people”	Notes
Scale of impact	30% (≈4,200 orders)	“[15:22] meera.i: impact - roughly 30% of checkout attempts failed between 14:05 and 14:40, about 4,200 failed orders. no data loss, the failed orders were never written”	Notes
Business or SLA impact	no SLA breach; 37 support tickets	“[15:25] ravi.kumar: no SLA breach, we're inside the monthly error budget. support got 37 tickets”	Notes
Data loss or exposure	none	“[15:22] meera.i: impact - roughly 30% of checkout attempts failed between 14:05 and 14:40, about 4,200 failed orders. no data loss, the failed orders were never written”	Notes
Incident start	14:05	“[14:07] priya.n: looking at the graphs it started ~14:05”	Notes
Detected	14:06	“[14:06] priya.n: seeing a wall of 500s on payments-api, checkout is failing for a lot of people”	Notes
Mitigated	14:40	“[14:40] ravi.kumar: rollback done, error rate dropping”	Notes
Resolved	15:10	“[15:10] ravi.kumar: all green for 30 min, closing. resolved 15:10”	Notes
Timezone	IST	“#inc-4821-payments (channel export, 3 Nov 2026, all times IST)”	Notes
How it was detected	user reported 500 errors on payments-api	“[14:06] priya.n: seeing a wall of 500s on payments-api, checkout is failing for a lot of people”	Notes
Monitoring gaps	deploy channel is too noisy, causing delay in linking deploy to errors	“took us 10 min to connect the deploy to the errors, the deploy channel is too noisy”	Notes
Fix applied	rollback of deploy 2291	“rollback done, error rate dropping”	Notes
Trigger	config change in deploy 2291 that set db pool max to 5	“deploy 2291 went out at 14:03, it included a config change for the connection pool”	Notes
Root cause	config change in deploy 2291 set db pool max to 5 instead of 50 (typo); CI doesn\'t validate config values	“root cause: the config change in deploy 2291 set the db pool max to 5 instead of 50 (typo). CI doesn\'t validate config values so it sailed through”	Notes

Field	Value	Supporting quote	Source
Affected services	payments-api	“[14:06] priya.n: seeing a wall of 500s on payments-api, checkout is failing for a lot of people”	Notes
Affected services	checkout-web	“[14:15] ravi.kumar: checkout-web is throwing too since it calls payments”	Notes
Affected services	payments-db	“[14:18] meera.i: db connections are maxed out on payments-db, the pool is saturated”	Notes
Mitigation steps	rolled back deploy 2291	“ok rolling back 2291”	Notes
People and teams involved	pri…	“pri… seeing a wall of 500s on payments-api, checkout is failing for a lot of people”	Notes
People and teams involved	ravi.kumar	“ravi.kumar: I'll take IC. calling it sev2 for now.”	Notes
People and teams involved	meera.i	“meera.i: db connections are maxed out on payments-db, the pool is saturated”	Notes
Contributing factors	CI doesn\'t validate config values	“root cause: the config change in deploy 2291 set the db pool max to 5 instead of 50 (typo). CI doesn\'t validate config values so it sailed through”	Notes
What went well	alert fired fast and the rollback was clean	“went well: alert fired fast and the rollback was clean”	Notes
What went wrong	took 10 min to connect the deploy to the errors due to noisy deploy channel	“went badly: took us 10 min to connect the deploy to the errors, the deploy channel is too noisy”	Notes